Annexure 1: Scope of Work

Design & Supply of telescopic belt conveyor for the Empty Bag facility unloading

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1. Background

We, YARA Fertilisers India Pvt Ltd, operates its Ammonia/ Urea complex of capacity 2000/ 3500 MTPD at Babrala, Sambhal District of Uttar Pradesh in Northern India which has a remarkable record in energy efficiency, reliability, safety, and environmental conservation.

The complex has one stream of Ammonia plant (capacity 2000 MTPD), two streams of Urea plant (capacity 1775 MTPD each) and at the downstream Packaging and dispatch (By rail and Road in 80/20% ratio) facility common for both urea plant streams.

2. Current process Description

We, YARA Fertilisers India Pvt Ltd, received the Empty bags from different suppliers through different type of Trucks. The Empty Bag lot is unloaded in the Empty Bag storage shed Manually through carrying a 50 Kg lot on head. During this unloading activity, the Workers has to pass through a Ramp & travels approx. 15 to 50 Mtrs for stacking.

The frequency of receipt & unloading of Empty Bag Truck is Minimum one per day.

3. Requirement:

Design and supply of a telescopic belt conveyor which is capable of handling 20T/h load. Bag lot/bundle will be shifted in the telescopic belt conveyor manually which will deliver bundle in the available belt conveyor.

4. Technical Inputs:

Description	Requirement	Remarks
Service	Handling of Empty Bag	Yes
Weight of Empty Bag's lot	50 Kg	Yes
Capacity of belt conveyor	20 Ton/Hr	Yes
Dimension of Lot/bundle, (LxBxH) mm	1000 X 700X250	Yes
Belt & Telescopic Conveyor Speed	0.6 Mtr/Sec	Yes (VFD make siemens/ABB/L&T)
Belt Width	800mm	Yes
Bag Placement of Belt	Length wise	Yes
No of Collection Points	1 No	Yes
No of Delivery Points	1 Nos	Yes
Extension of telescopic conveyor	9meter*	Total Length 14-meter, Fixed Length 5-meter, Flexible length 3+3+3Meter.

Support	Hinged at one end and other end movable (up and down)	Maximum Inclination should we provide to conveyor is **1200mm.**
Safety Devices		
Manual Stop switch at least three position	Required	Teknic make Contact rating-5Amp, IP65. 3 no's Emergency switch we provide.
Mechanical limit switch	1no-electric operated	Teknic -make Heavy duty 2NO+2NC Contact rating5Amp, IP65
Proximity Sensor for telescopic Conveyor	Capacitance type-2nos	2out of 1 logic with safe distance sensing at front. (Omron make)
Others		
Belt Specifications	Natural / Synthetic Rubber, M24 315/3, Min top Cover thk:3mm & for Bottom:1.5 mm	Yes
Material of structure	As per design	Approval to be taken, MOC of each component required in technical bidding
Driven Motor Make	BBL	
Panel for operation	As per design	Min-IP65,Ammeter, indication for each motion, with proper marking switch, with cannopy
Gear Box Make	Bonfiglioli	Yes
Power	All component should energy efficient	Yes
Duty	Intermittent	Dependent truck frequency, Min 2 truck per day
Clearance	Minimum clearance 100mm from platform	Truck platform height is 1350mm
Design	Design basis and calculation approval will be taken after getting order	Yara will ensure for the minimum strength and power

**Preference will be given to minimum static (folded) length conveyor*.

- **Electricals**

Sr. No.	COMPONENTS	MAKE
1	Electric Motor	BBL
2	Contactor/Over Load Relays/VFD	Siemens /ABB/L&T
3	MCB/MPCB/Indicators	Siemens
4	Panel Box Enclosure	DL TECH
5	Push Buttons/Limit Switches	Teknik
6	Fittings	Standard
7	Cables	Polycab/Avanti

Hydraulics

Sr. No.	COMPONENTS	MAKE
1	Power Pack	Target innovation
2	Hydraulic Cylinders	DL TECH
3	Hydraulic hose pipe	Parker

Others

Sr. No.	COMPONENTS	MAKE
1	Bearing & Bearing Units	SKF
2	Return Roller	DL TECH Make
3	Drive & Driven Roller	DL TECH Make
4	Chain & Sprockets	TI Diamond
5	MOC	Mild Steel Jindal/Apollo
6	Surface Treatment	Powder Coated

5. Scope of Work

The Bidder's scope of work shall be the complete Detailed Engineering, Procurement of Materials , Manufacturing , testing & supply of telescopic Conveyor with required accessories.

The scope of supply includes but not limited to the followings-

5.1 Design of telescopic Conveyor as per YARA specification sheet & submission design & GA drawing for submission of same for YARA approval - **Yes**

5.2 Procurement of all material required for Telescopic conveyor. -**Yes**

5.3 Establishment of Quality Assurance Plan & submit for YARA approval- **Yes**
5.4 Procurement & Fabrication as per finalized Quality Assurance Plan-**Yes**
5.5 Trial of complete Telescopic Conveyor at Bidder works- **Yes**
5.6 Supply of complete telescopic conveyor at site. - **Yes**
5.7 Submission of all the As-built drawings, Data sheets & QAP adherence with necessary documents for YARA final records. -**Yes**
5.8 Submission of Project Schedule, Monthly Progress Report. - **Yes**

Note: Bidder is suggested to Maximum utilization of modern technologies after YARA approval

6. Special requirements:

6.1 **Pulleys:**

6.1.1 All drum pulleys (Head & Tail Pulley) are to be provided with XT hub & bushing/ shrink fit instead of conventional (only keyed hub) pulley**.- Yes**

6.1.2 Head Pulley shall have 12mm thk Natural Rubber lagging Diamond pattern groove, hot vulcanized method**,- Yes**

6.1.3 Tail Pulley shall have min 8mm thick Natural Rubber plain lagging, hot vulcanized method,- **Yes**

6.1.4 Pulley runout shall be within +/-0.5% of the nominal O.D (without lagging)-**Yes**

6.1.5 Pulley shall be statically balanced- **Yes**

6.1.6 Pulley to be Punch Marked-**Yes**

6.1.7 Keyway as IS 2048- **Yes**

6.1.8 All weld are min, 6 mm fillet unless or otherwise mentioned- **Yes**

6.1.9 Shell plates shall have single longitudinal continuous weld -**Yes**

6.1.10 Shor hardness shall be 60°+/-5° on Scale A- **Yes**

6.2 **Idler Units**

6.2.1 Each roller for idlers shall consist of a pair of anti-friction bearings of ample capacity and shall have minimum double labyrinth seals for keeping the lubricant in and dirt out. The bearings and sealing devices shall be mounted in the housing with suitable rain cap to prevent water entering inside the housing. The bearing housing shall be welded on either side of the rollers with through shafts for proper load support. – **(bearing housing is bolted on side of conveyor**)

6.2.2 The bearing housings shall be minimum 3 mm thick and shall be made of pressed steel, welded to the idler rollers simultaneously. The maximum co-efficient of friction for idler unit shall not exceed 0.015.- **No**

6.2.3 The carrying idler units shall be heavy duty suitable for continuous operation. – **Yes**

6.2.4 Return idlers shall be mounted on sturdy steel or malleable iron brackets bolted to the underside of stringers. Return idlers spacing near the head and tail pulleys shall be less than 2 meters.- **Yes**

6.2.5 The Sharp corner of Idler bracket shall be round off & having slots in place of bolt holes to align the belt during running conditions. – **Yes**

6.2.6 All carrying & return idlers of shall be rubber lagged. - **Yes**

6.2.7 All bearings and mounting should be SKF make only -**Yes**

6.2.8 A common greasing position for the all bearings to be provided- **Yes**

6.3 **Structural Details**

6.3.1 All structural steel material shall conform to IS 2062. All shop connections and field connections shall be welded or bolted type. Field connections in equipment supports and equipment platforms shall be with friction grip high strength bolts or welded connections. - **Yes**

6.3.2 Civil foundation detail and drawings to be provided for the Telescopic conveyor- **Yes**

6.4 **Motor Details**

1.1.1 Terminal box should be metallic.- **Yes**

1.1.2 Motor Insulation must be at least F class-**Yes**

1.1.3 Efficiency Class: IE2/IE3 – **IE2**

1.1.4 Motor Duty is continuous rated i.e. S1- **Yes**

1.1.5 IP class must be IP55 (Ingress Protection)- **No**

1.1.6 Motor Cooling fan should be metallic.- **Yes**

1.1.7 TEFC Motor, suitable up to 50° C ambient temp. **- Yes**

1.1.8 3 Phase; 415 V ± 10%; 50 Hz ± 5% -**Yes**

1.1.9 Motor Terminal box for New BC 16 should be at the top & the cable entry should be from walkway platform side. The Terminal box should be suitable foe 3Cx10 Sqmm.- **Yes**

1.1.10 For portable conveyors, Motor & Gear box should be parrel to the conveyor. Motor Terminal box should be according to Cable Entry form open area.-**Yes**

6.5 **Others**

6.5.1 The vibration limit shall be as per zones A and B of ISO 10816-1.**- Yes**

6.5.2 Noise levels produced by any rotating equipment individually and collectively shall not exceed 85 dB (A) measured at distance of 1.5 Mtrs from the source in any direction under any load condition.- **Yes**

6.5.3 Idlers shall be with grease packed sealed deep grove anti-seize type ball bearing. Bearings shall have multi labyrinth seals to protect from dust**.- Yes**

6.5.4 All the Fasteners shall be Unbroko make with two washers.- **Yes**

2. Painting:

2.1 All CS external surfaces of shop fabricated equipment/item shall be painted as per the following.

2.2 Surface Preparation: Grit blasting to SA 2-1/2 Grade. **-No**

2.3 Total DFT of painting shall be 300µ (including primer, intermediate & finish paint) & the final coat shall be Red colour. - **Yes**

Note:

The paint specification shall be compatible with the external environment consisting of fine urea dust particles which is very corrosive and hygroscopic in nature. -**Yes**

3. Minimum Inspection Requirements:

3.1 **Idler Assembly**

3.1.1 TC verification of Raw materials (MS tubes, shafts, bearings, Brackets etc.). **- Yes**

3.1.2 Welding document (including NDT) review. **- Yes**

3.1.3 Final inspection of Brackets. - **Yes**

3.1.4 Round-off edges of Brackets. **- Yes**

3.2 **Idler Roller**

3.2.1 Dimensions of rollers, quality, and radial run out checks.**- Yes**

3.2.2 Dust ingression test- **No**

3.2.3 Water ingression test. **- No**

3.2.4 Friction Coefficient tests. - **No**

3.2.5 Shore hardness of rubber discs (for impact idlers).- **No**

3.2.6 Open inspection of bearing and rollers (at random).**- Yes**

3.3 **Idler with bracket**

3.3.1 Dimensional & troughing angle measurement of brackets.**- Yes**

3.3.2 Free movements of rollers**.- Yes**

3.3.3 Gap between roller edges.-**Yes**

3.3.4 Level of rollers.- **Yes**

3.3.5 Verification of thrust bearing, swiveling of self-aligning idlers about the pivot.- **Yes**

3.4 **Pulleys**

3.4.1 Raw material TC verification (Pulley shell, shaft, hubs, end discs, Plummer blocks, bearing etc.) - **Yes**

3.4.2 Welding document (including NDT) review. - **Yes**

3.4.3 Inspection as per drawing, dimensional measurements. - **Yes**

3.4.4 Lagging rubber examination and verification of hardness. -**Yes**

3.4.5 Lagging rubber peel off test on random samples. -**Yes**

3.4.6 Static balancing after rubber lagging. - **Yes**

3.4.7 Verification of bearings. - **Yes**

3.4.8 Examination of hub and shaft fixing arrangement. - **Yes**

3.4.9 UT for shaft diameter>50mm. **– Yes**

3.5 **Gear Box:**

3.5.1 Gear Blanks, Shafts, Bearings, Pinions & Housing: MTC, 100% UT of forging**, - Yes**

3.5.2 Gears: Heat Treatment Chart Review, Hardness Check, Case depth for hardened and ground gears, PT on gear tooth**- Yes**

3.5.3 Performance test (No Load): Vibration Check, b) Noise level, c) Verification of Temperature Rise of Oil Bath d) Verification of speed ratio, e) Dimensional Inspection- **Yes**

3.6 **Structural Items:**

3.6.1 Raw material TC verification. **- Yes**

3.6.2 Welding document (including NDT) review. - **Yes**

3.6.3 Final inspection assembly- Visual, dimensional. - **Yes**

3.7 **Painting**

3.7.1 Surface Finish measurement- **No**

3.7.2 DFT measurement- **No**

3.8 **Electrical Motors**

3.8.1 Performance Test Certificate- **Yes**

4. Expected Completion Time

The delivery of above assignment shall be in a 2 months' time frame from the date of award of contract. - **Yes**

5. Applicable Engineering Codes & Standards:

It is the primary responsibility of vendor (Successful Bidder) to design & construct all items related to this contract in accordance with the latest industry standard & codes, including all supplements and addendums issued at time of the order. The applicable codes shall include but not limited to the followings.

IS 11592: Selection & Design of Belt Conveyor. **- NO**
CEMA: Belt Conveyors for bulk materials. - **NO**
IS 1891: Conveyor & Elevator Textile Belting. - **No**
IS 8531: Pulleys for Belt Conveyor- **No**
IS 8598: Idlers & Idlers sets for Belt Conveyor. - **No**
IS 9295: Steel Tubes for Idlers for Belt Conveyor.- **No**
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6. Preservation, protection, Packing & Dispatch:

6.1 Cleaning, preservation, storage, packing and preparation for Loading/dispatch of the equipment, equipment components shall be performed according to the manufacturer's standard/good engineering practices. -**Yes**

6.2 All spares parts shall be packed separately in wooden packet to avoid damage in transport. -**Yes**

6.3 All commissioning and 2 years operational spares shall be packed and marked separately to avoid mix up at site**.(Spare part is not provide in same offer, extra cost will be applicable for spare part).**

7. LD Clause:

In the event, the supplier fails to deliver the equipment/components within the contractual delivery date, then the supplier shall pay to YARA as and by way of Liquidated Damages of 0.5% (Point Five Percent) of the basic value for each week delay or part thereof, subject to maximum of 5% (Five Percent) of basic value. - **Yes**

8. Guarantee/Warrantee:

8.1 Vendor (i.e. successful bidder) shall guarantee all components related to this contract including spare materials against faulty design, improper material of construction and poor workmanship in addition to performance guarantee. Approval or reviewing by YARA for design calculation and detailed shop drawing, will not in any way absolve the vendor from his responsibility. Should any repair or replacement be necessary owing to any type of failure on account of design, material and workmanship of the item, the vendor shall in view of this guarantee be bound to replace the same either in part or whole without any time & cost implication to the purchaser. Repaired or replaced part shall also be covered by same guarantee as in case of main supply. - **Yes**

8.2 Guarantee/warrantee shall be valid for 18 months from date of dispatch or 12 months from date of commissioning whichever is earlier. - **Yes**

8.3 **Performance Guarantees:**

8.3.1 Telescopic Conveyor and belt feeder shall run centrally without any misalignment. -**Yes**

8.3.2 The vibration limit of conveyor drives as measured at center line of pulley bearings, motor bearings and gear box bearings shall be as per zones A and B of ISO 10816-1.- **Yes**

8.3.3 The noise level at a distance of 1.5 Mtrs from the drive of conveyor shall be less than 85 dB (A). - **Yes**

8.3.4 The vibrations in structural supports of conveyor structures shall be less than that specified. - **Yes**

9. Quality Assurance & Quality Control:

9.1 Quality Assurance (QA) shall mean the organization set up, procedures as well as test methods and facilities developed by the vendor/supplier in order to assure that all equipment / items leaving vendor's shop are of the highest possible quality that is either equal to or better than the requirements specified.

9.2 Quality Control (QC), shall mean all the tests, measurements, checks and calibration which are to be carried out at vendor's shop in order to compare the actual characteristics of the equipment/unit/system with the specified ones, along with furnishing of the relevant documentation (certificates/ records) containing the data or result of these activities.

9.3 Vendor (Successful Bidder) shall submit a comprehensive description (manual) of QA/QC measures contemplated by them for implementation regarding this specification. It is contractual obligation of the vendor to develop and implement adequate QA/QC systems.

9.4 QA/QC system shall cover all products and services of the contract i.e. documentation material, shop and site fabrication, transportation and site works, including job subcontracted by the vendor.

9.5 Bidder shall permit the YARA appointed Quality personal at their premises for inspection. YARA shall also communicate the visit plan of their Nominated Quality cum expediter person.

10. Submission of Proposal & Bidding Instructions

10.1 Before submitting the proposal, plant visits/online discussion may be made by the Bidder to understand the system requirements, clarifications, and collection of any other data from the site.

10.2 Bidder's proposal at offer submission stage shall contain the

10.2.1 Bidder's organization profile & experience of similar jobs in the last 5 years

10.2.2 Detailed list of modification on the existing facilities

10.2.3 Tentative schedule with bar chart and detailed activity descriptions at various stages

10.2.4 Manpower organization chart of the bidder pertaining to the scope of this contract with responsibilities specified.

10.2.5 Quality control program as necessary.

10.2.6 List of exclusions/deviations.

10.3 The Technical Bids along with all required attachments & un-priced commercial bid should be sent through E mail to the following address on or before by 30th Dec'2021.

Email: saurabh.mathur@yara.com

10.4 The guidelines for submission of Priced Commercial proposal shall be sent separately after completion of Technical & Un-priced commercial bid clarifications. The Bid closing date for priced offer shall be communicate accordingly.

11. General Terms & Communications

11.1 All bids shall be submitted in accordance with instructions contained herein in sealed covers duly subscribed with enquiry no. & due date.

11.2 Bid shall be in total conformity with technical specifications and conditions of service/purchase, if not, exclusions / deviations shall be separately listed out & given with technical & un-priced commercial bid. Without any such specific stipulation, bidder's acceptance of specifications and conditions of purchase shall be taken for granted.

11.3 Notwithstanding anything to the contrary contained herein and irrespective of what have been stated in the bidding document enclosed, owner reserves his absolute right to accept or reject the bid, at his sole discretion without assigning any reason whatsoever.

11.4 *Confidentiality:* All information which is either provided by and/or obtained from YARA Fertilisers India Pvt Ltd ("YARA"), whether or not the same has been designated a "Confidential" shall be regarded and kept as confidential and be limited to those representatives who have a need to know the same. Bidder agrees not to use the name of YARA or disclose the existence of the Enquiry/Purchase order/ Contract/Agreement in any advertising, promotion or other written or oral disclosure without the prior written consent of YARA.

12. Enclosures

12.1 Concept layout for Empty Bag facility: Assessment of Telescopic Conveyor Height & length- **We can cover upper portion instead of full enclosure (cost not include in the offer).**